

Correction of the final exam

Exercise 1: (4 points)

1. True or false and negation

The statement " $\exists x \in \mathbb{R}, x^2 - 5 \geq 0$ " is True.

Justification: $\exists x = 6 \in \mathbb{R}, x^2 - 5 = 6^2 - 5 > 0$.

Negation: $\forall x \in \mathbb{R}, x^2 - 5 < 0$.

2. Proof by contrapositive that $\forall m \in \mathbb{Z}$:

$$(m^2 - 3 \text{ is odd}) \Rightarrow (m \text{ is even}).$$

By contrapositive, we prove that $\forall m \in \mathbb{Z}$:

$$(m \text{ is odd}) \Rightarrow (m^2 - 3 \text{ is even}).$$

Let $m \in \mathbb{Z}$.

$$m \text{ is odd} \Leftrightarrow \exists k \in \mathbb{Z}, m = 2k + 1.$$

$$\text{Then, } m^2 - 3 = (2k + 1)^2 - 3 = (2k)^2 + (1)^2 + 2(2k)(1) - 3$$

$$= 4k^2 + 1 + 4k - 3$$

$$= 4k^2 + 4k - 2$$

$$= 2(2k^2 + 2k - 1)$$

$$= 2M. \quad (M = 2k^2 + 2k - 1 \in \mathbb{Z})$$

Then $m^2 - 3$ is even

Therefore, $\forall m \in \mathbb{Z}, (m^2 - 3) \text{ is odd} \Rightarrow m \text{ is even}$

0,5

1

0,5

0,5

1,5

P₁

Exercise 2: (7pts)

I. Let R be a relation defined on $\mathbb{Z}$ as follows:

$$a R b \Leftrightarrow \exists n \in \mathbb{N}, a - b = 3n$$

1. Verify if: $(-6) R (-3)$, $(-3) R (-6)$

$$(-6) - (-3) = -6 + 3 = -3 = 3 \times (-1), \text{ then } (-6) \not R (-3).$$

$$(-3) - (-6) = -3 + 6 = 3 = 3 \times 1, \text{ then } (-3) R (-6).$$

2. Show that R is an order relation

$$(R \text{ is an order relation}) \Leftrightarrow \begin{pmatrix} R \text{ is reflexive,} \\ \text{antisymmetric,} \\ \text{and transitive.} \end{pmatrix}$$

① Reflexivity

$$R \text{ is reflexive} \Leftrightarrow \forall a \in \mathbb{Z}, a R a$$

Let $a \in \mathbb{Z}$.

$$a - a = 0 = 3 \times 0. \text{ Then } a R a$$

Therefore R is reflexive.

② Antisymmetry

$$R \text{ is antisymmetric} \Leftrightarrow \forall a, b \in \mathbb{Z}, (a R b) \wedge (b R a) \Rightarrow a = b$$

Let $a, b \in \mathbb{Z}$.

$$\begin{cases} a R b \\ b R a \end{cases} \Rightarrow \begin{cases} \exists n_1 \in \mathbb{N}, a - b = 3n_1 \\ \exists n_2 \in \mathbb{N}, b - a = 3n_2 \end{cases}$$

$$\Rightarrow \exists n_1, n_2 \in \mathbb{N}, (a - b) + (b - a) = 3n_1 + 3n_2$$

$$\Rightarrow \exists n_1, n_2 \in \mathbb{N}, 3(n_1 + n_2) = 0$$

$$\Rightarrow n_1 = 0 \wedge n_2 = 0 \Rightarrow a = b$$

Then R is antisymmetric.

③ Transitivity

R is transitive $\Leftrightarrow$

$$\forall a, b, c \in \mathbb{Z}, \\ (a R b) \wedge (b R c) \Rightarrow (a R c)$$

Let $a, b, c \in \mathbb{Z}$.

$$\begin{cases} a R b \\ \wedge \\ b R c \end{cases} \Rightarrow \begin{cases} \exists n_3 \in \mathbb{N}, a - b = 3n_3 \\ \exists n_4 \in \mathbb{N}, b - c = 3n_4 \end{cases}$$

$$\Rightarrow \exists n_3, n_4 \in \mathbb{N}, (a - b) + (b - c) = 3n_3 + 3n_4$$

$$\Rightarrow \exists n_3, n_4 \in \mathbb{N}, a - c = 3(n_3 + n_4)$$

$$\Rightarrow \exists n = n_3 + n_4 \in \mathbb{N}, a - c = 3n$$

$$\Rightarrow a R c$$

Therefore R is transitive.

Conclusion:

R is an order relation

3. Is the order total?

The order is total $\Leftrightarrow$

$$\forall a, b \in \mathbb{Z}, \\ (a R b) \vee (b R a)$$

The order is not total, because:

$$\exists a = 2, b = 1 \in \mathbb{Z}, 2 \not R 1 \text{ and } 1 \not R 2.$$

$$(\forall n \in \mathbb{N}, 2 - 1 = 1 \neq 3n \text{ and } \forall n \in \mathbb{N}, 1 - 2 \neq 3n)$$

II. Let f be a function such that: $f: [-1, 3] \rightarrow [0, 2]$
 $x \mapsto \frac{x^2}{x+2}$

1. Determine $f(\{-1, 0, 2\})$ and $f^{-1}(\{\frac{1}{3}, 2\})$

$$\bullet f(\{-1, 0, 2\}) = \{f(x) \in [0, 2], x \in \{-1, 0, 2\}\}$$

$$f(-1) = \frac{(-1)^2}{-1+2} = \frac{1}{1} = 1 \in [0, 2]$$

$$f(0) = \frac{0}{0+2} = 0 \in [0, 2], f(2) = \frac{2^2}{2+2} = \frac{4}{4} = 1 \in [0, 2]$$

$$\text{Then } f(\{-1, 0, 2\}) = \{0, 1\}.$$

$$\bullet f^{-1}(\{\frac{1}{3}, 2\}) = \{x \in [-1, 3], f(x) \in \{\frac{1}{3}, 2\}\}$$

$$f(x) = \frac{1}{3} \Leftrightarrow \frac{x^2}{x+2} = \frac{1}{3} \Leftrightarrow 3x^2 = x+2 \Leftrightarrow 3x^2 - x - 2 = 0$$

$$\Leftrightarrow x = \frac{-2}{3} \vee x = 1$$

$$\begin{matrix} \in [-1, 3] & \in [-1, 3] \end{matrix}$$

$$f(x) = 2 \Leftrightarrow \frac{x^2}{x+2} = 2 \Leftrightarrow x^2 = 2x+4 \Leftrightarrow x^2 - 2x - 4 = 0$$

$$\Leftrightarrow x = 1 - \sqrt{5} \vee x = 1 + \sqrt{5}$$

$$\begin{matrix} \notin [-1, 3] & \notin [-1, 3] \end{matrix}$$

Then

$$f^{-1}(\{\frac{1}{3}, 2\}) = \{\frac{-2}{3}, 1\}.$$

2. Is f injective? Surjective? Bijective? Justify.

$\bullet f$ is not injective, because:

$$\exists x_1 = -1, x_2 = 2 \in [-1, 3], f(-1) = f(2) = 1 \text{ and } -1 \neq 2$$

$\bullet f$ is not surjective, because:

$$\exists y = 2 \in [0, 2], \forall x \in [-1, 3], y \neq f(x).$$

$\bullet f$ is not bijective, because: f is not injective
 (or: f is not surjective)

Exercise 3: (4 points)

$$\text{Let } k = \frac{2i + 2\sqrt{3}}{4i} \in \mathbb{C}.$$

1. Write k in algebraic form ($x+iy$)

$$\begin{aligned} k &= \frac{2i + 2\sqrt{3}}{4i} = \frac{(2i + 2\sqrt{3})(-4i)}{(4i)(-4i)} \\ &= \frac{8 - i8\sqrt{3}}{16} = \frac{8}{16} - i \frac{8\sqrt{3}}{16} = \frac{1}{2} - i \frac{\sqrt{3}}{2} \end{aligned}$$

2. The modulus r of k :

$$r = |k| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = \sqrt{1} = 1.$$

• An argument $\theta \in [0, 2\pi]$ for k :

$$\begin{cases} \cos \theta = \frac{1}{2} \\ \sin \theta = -\frac{\sqrt{3}}{2} \end{cases} \Rightarrow \theta = \frac{5\pi}{3} \in [0, 2\pi]$$

• The exponential form of k is: $k = e^{i\frac{5\pi}{3}}$

3. Use De Moivre's formula to find $k^3 \in \mathbb{R}$.

$$k^3 = \left(e^{i\frac{5\pi}{3}}\right)^3 = e^{i5\pi} = -1$$

4. Solve in $\mathbb{C}$ the equation $z^3 - k^3 = 0$.

$$z^3 - k^3 = 0 \Leftrightarrow z^3 - (-1) = 0 \Leftrightarrow z^3 + 1 = 0 \Leftrightarrow z^3 = -1$$

We have to find the cube roots of (-1) . Use the formula

$$z_k = r^{\frac{1}{n}} e^{i\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right)}, (r=|-1|, \theta=\text{Arg}(-1), n=3, k=0,1,2)$$

$$z_0 = 1^{\frac{1}{3}} e^{i\left(\frac{\pi}{3} + 0\right)} = e^{i\frac{\pi}{3}} = \boxed{\frac{1}{2} + i\frac{\sqrt{3}}{2}}$$

$$z_1 = e^{i\left(\frac{\pi}{3} + \frac{2\pi}{3}\right)} = \boxed{-1}, \quad z_2 = e^{i\left(\frac{\pi}{3} + \frac{4\pi}{3}\right)} = e^{i\frac{5\pi}{3}} = \boxed{\frac{1}{2} - i\frac{\sqrt{3}}{2}}$$

0,5

0,5

0,5

0,5

0,5

1,5

P.5

Exercise 4:

5 points

I. Let $E = \{(x, y, z) \in \mathbb{R}^3, \frac{1}{2}xy = z\} \subset \mathbb{R}^3$.

$$u = (1, 4, 2), \quad v = (3, -2, -3) \in \mathbb{R}^3.$$

1). Is $u \in E$? $v \in E$?

$$\frac{1}{2} \times 1 \times 4 = 2 \Rightarrow u \in E.$$

$$\frac{1}{2} \times (3) \times (-2) = \frac{-6}{2} = -3 \Rightarrow v \in E.$$

• Is $u+v \in E$

$$u+v = (4, 2, -1)$$

$$\frac{1}{2} \times 4 \times 2 = \frac{6}{2} = 3 \neq -1 \Rightarrow u+v \notin E.$$

2) Is E a subspace of $\mathbb{R}^3$?

E is not a subspace of $\mathbb{R}^3$ because it is not closed under addition. That's because

$$\exists u = (1, 4, 2), v = (3, -2, -3) \in E \text{ but } u+v \notin E.$$

II) Let $F = \{(x, y) \in \mathbb{R}^2, x - 2y = 0\} \subset \mathbb{R}^2$.

Prove that F is a subspace of $\mathbb{R}^2$.

① $0_{\mathbb{R}^2} = (0, 0) \in F$, because $0 - 2 \times 0 = 0$.

② Let $u = (x, y), v = (x', y') \in F$, then $x - 2y = 0$ and $x' - 2y' = 0$.

$$u+v = (x+x', y+y') = (X, Y)$$

$$X - 2Y = (x+x') - 2(y+y') = x - 2y + x' - 2y' = 0$$

Then $u+v \in F$

③ Let $u = (x, y) \in F$ ($x - 2y = 0$) and let $\lambda \in \mathbb{R}$.

$$\lambda u = (\lambda x, \lambda y) = (X, Y).$$

$$X - 2Y = \lambda x - 2\lambda y = \lambda(x - 2y) = \lambda \times 0 = 0$$

Then $\lambda u \in F$. Conclusion: F is a subspace of $\mathbb{R}^2$.

0,5

0,5

0,5

1

0,5

1

1